Challenge Problem Set 6

Math 292 Spring 2026
Rutgers University

Exercise 1:

Suppose A is a self-adjoint matrix with non-positive eigenvalues.

Then, by the Spectral Theorem, A can be diagonalized into an orthonormal basis. That is

$$A = UDU^*,$$

where U is a unitary matrix and D is a diagonal matrix.

Thus, we also have

$$e^{tA} = Ue^{tD}U^*.$$

For any $\mathbf{x}$, we can rewrite it as a linear combination of this orthonormal basis. That is, for constants $\mathbf{c}$, we have

$$\begin{aligned} ||\mathbf{x}||^2 &= ||U\mathbf{c}||^2 \\ &=< U\mathbf{c}, U\mathbf{c} > \\ &= ||\mathbf{c}||^2 \end{aligned}$$

Let A have n columns. Then

$$||\mathbf{x}||^2 = \sum_{i=1}^{n} c_i^2.$$

Now we consider

$$\begin{aligned} ||e^{tA}\mathbf{x}||^2 &= ||Ue^{tD}U^*\mathbf{x}||^2 \\ &= ||Ue^{tD}\mathbf{c}||^2 \\ &=< e^{tD}\mathbf{c}, e^{tD}\mathbf{c} > \\ &= \sum_{i=1}^{n} e^{2t\lambda_i} c_i^2. \end{aligned}$$

Since $t \geq 0$ and $\lambda_i \leq 0$, it follows that $t\lambda_i \leq 0$ and $e^{2t\lambda_i} \leq e^0 = 1$. Thus,

$$||e^{tA}\mathbf{x}||^2 \leq \sum_{i=1}^{n} c_i^2 = ||\mathbf{x}||^2.$$

Taking the square root of both sides gives

$$||e^{tA}\mathbf{x}|| \leq ||\mathbf{x}||.$$

Therefore, we conclude that e^{tA} is contractive if A has all non-positive eigenvalues.

Exercise 2:

Suppose K is a skew-adjoint square matrix. Then, we compute

$$(iK)^* = -i(-K) = iK.$$

That is, iK is a self-adjoint matrix.

Thus, by Spectral Theorem, the eigenvalues of K must be purely imaginary and K can be diagonalized into an orthonormal basis UDU^*. Thus, $e^{tK} = Ue^{tD}U^*$. Consider the adjoint of e^{tK}:

$$\begin{aligned}(e^{tK})^* &= (Ue^{tD}U^*)^* \\ &= Ue^{-tD}U^* \\ &= e^{-tK}.\end{aligned}$$

We can check that this is exactly the inverse:

$$e^{tK}(e^{tK})^* = e^{tK}e^{-tK} = I = e^{tK}(e^{tK})^{-1}.$$

Thus, we have that e^{tK} is unitary.

To show e^{tK} is also norm-preserving, we compute

$$\begin{aligned}||e^{tK}\mathbf{x}||^2 &=< e^{tK}\mathbf{x}, e^{tK}\mathbf{x} > \\ &= \mathbf{x}^T(e^{tK})^*e^{tK}\mathbf{x} \\ &= \mathbf{x}^T I\mathbf{x} \\ &= \mathbf{x}^T\mathbf{x} \\ &= ||\mathbf{x}||^2.\end{aligned}$$

Taking the square root of both sides gives $||e^{tK}\mathbf{x}|| = ||\mathbf{x}||$. Hence e^{tK} is norm-preserving.

Exercise 3:

Let $A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*) =: H + K$.

Suppose H has non-positive eigenvalues.

Taking the exponential of $A = H + K$, we have

$$e^{tA} = e^{t(H+K)}.$$

Then, applying the trotter product, we get

$$e^{t(H+K)} = \lim_{n\to\infty}(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n.$$

Since $T_n \to T \implies ||T_n|| \to ||T||$ under the operator norm, we have

$$\begin{aligned}||e^{t(H+K)}|| &= ||\lim_{n\to\infty}(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n|| \\ &= \lim_{n\to\infty}||(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n||\end{aligned}$$

Consider $(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n$, because $||AB|| \leq ||A||\ ||B||$ under the operator norm, we have

$$\begin{aligned} ||(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n|| &\leq ||(e^{\frac{t}{n}H}e^{\frac{t}{n}K})||^n \\ &\leq ||e^{\frac{t}{n}H}||^n\ ||e^{\frac{t}{n}K}||^n \end{aligned}$$

Since H is also self-adjoint, it follows that $||e^{tH}|| \leq 1$. Since K is skew-adjoint, it follows that $||e^{tK}|| = 1$.

Thus,

$$||e^{\frac{t}{n}H}||^n\ ||e^{\frac{t}{n}K}||^n \leq 1^n \cdot 1^n = 1.$$

Hence,

$$||e^{tA}|| = \lim_{n\to\infty} ||(e^{\frac{t}{n}H}e^{\frac{t}{n}K})^n|| \leq \lim_{n\to\infty} 1 = 1.$$

That is, $||e^{tA}|| \leq 1$.

Therefore, we conclude that e^{tA} is contractive if H has only non-positive real eigenvalues.

Exercise 4:

Suppose that A is any constant square matrix and that e^{tA} is contractive for all $t \geq 0$. By definition of contractive, we have

$$\begin{aligned} ||e^{tA}x||^2 &\leq ||x||^2 \\ ||e^{tA}x||^2 - ||x||^2 &\leq 0 \end{aligned}$$

Notice that this is equal to $||e^{tA}x||^2 - ||e^0x||^2 \leq 0$ since $e^0 = I$. Then, dividing through by t and taking $t \to 0$, we have

$$\lim_{t\to 0} \frac{||e^{tA}x||^2 - ||x||^2}{t} \leq 0.$$

Notice this is exactly the definition of the derivative. So,

$$\frac{d}{dt}||e^{tA}x||^2\ \Big|_{t=0} \leq 0.$$

Computing the derivative of $||e^{tA}x||$, we have

$$\frac{d}{dt}||e^{tA}x||^2 = \langle Ae^{tA}x, e^{tA}x\rangle + \langle e^{tA}x, Ae^{tA}x\rangle.$$

Evaluating this at $t = 0$ gives

$$\begin{aligned} \frac{d}{dt}||e^{tA}x||^2\ \Big|_{t=0} &= \langle Ax, x\rangle + \langle x, Ax\rangle \\ &= 2Re\langle x, Ax\rangle. \end{aligned}$$

Using the previous equation $\frac{d}{dt}||e^{tA}x||^2\ \Big|_{t=0} \leq 0$, this reduces to

$$Re(x, Ax) \leq 0$$

Therefore, we conclude that A is dissipative.

Exercise 5:

Consider the following matrices:

$$A_1 = \begin{bmatrix} -\frac{1}{3} & 1 \\ 0 & -\frac{1}{3} \end{bmatrix} \quad A_2 = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$$

Notice that A_1 has eigenvalues $\lambda_1 = \lambda_2 = -\frac{1}{3}$ and A_2 has eigenvalues $\mu_1 = \mu_2 = -1$.

Consider the associated self-adjoint part of A_1, that being H_1.

$$\begin{aligned} H_1 &= \frac{1}{2}(A_1 + A_1^*) \\ &= \frac{1}{6}\begin{bmatrix} -2 & 3 \\ 3 & -2 \end{bmatrix} \end{aligned}$$

Then, we solve for its determinant:

$$det(H_1) = -\frac{5}{36} < 0.$$

Recall that $det(H_1) = \lambda_1 \cdot \lambda_2$. Thus, H_1 has two eigenvalues of opposite signs. From exercises 3 and 4, we concluded that H has negative eigenvalues if and only if A is dissipative. Since H_1 has a positive eigenvalue, it follows that A_1 isn't dissipative. It follows that e^{tA_1} is not contractive as well.

Next, we consider the associated self-adjoint part of A_2, that being H_2.

$$\begin{aligned} H_2 &= \frac{1}{2}(A_2 + A_2^*) \\ &= \begin{bmatrix} -1 & \frac{1}{2} \\ \frac{1}{2} & -1 \end{bmatrix}. \end{aligned}$$

Then, solving for its determinant:

$$det(H_2) = \frac{3}{4} > 0.$$

So H_2's eigenvalues are of the same sign. Notice that $tr(H_2) = -2$ is negative. Hence, both eigenvalues must also be negative. Since H_2 has only non-positive eigenvalues, we conclude that A_2 is dissipative, and so e^{tA_2} is contractive.